UConn Private Container Registry (Quay) — Getting Started

For OpenShift application teams (regular users)

Registry URL: https://quay.apps.uconn.edu

Sign-in: Use “Sign in with UConn EntraID” and your UConn email address.

1) What you get

- A team organization in Quay (same name as your team/namespace).
- Three standard repositories (created by admins): dev, test, prod.
- Robot accounts for pushing, scoped per repo/environment (e.g., <org>+dev can push to <org>/dev).
- An OpenShift imagePullSecret is also created by automation for cluster pulls (ask the platform team if you don’t see it).

2) Browse and verify access in the UI

- Go to https://quay.apps.uconn.edu and sign in with EntraID (regular UConn email).
- Select your organization (e.g., its-ssg).
- You should see repositories: dev, test, prod.

Note: Regular users typically cannot view robot accounts in the UI; admins manage robot credentials.

3) Install a container client

- Podman (preferred): https://podman.io/getting-started/installation (dnf install @container-tools -y)
- Docker: https://docs.docker.com/get-docker/

4) Get robot credentials

To push images, you’ll use a robot account username and token. These are provided by your org admins / Platform Engineering.

5) Login from the command line

Use the robot username + token:

```
podman login quay.apps.uconn.edu

# Username: <org>+dev   (example: its-ssg+dev)

# Password: <robot_token> (provided by admin)
```

6) Tag and push (recommended workflow)

Push to the environment repo you are targeting:

```
# Example: push to dev

podman tag myapp:1.0 quay.apps.uconn.edu/<org>/dev:myapp_v1.0

podman push quay.apps.uconn.edu/<org>/dev:my_app_v1.0


# Later promote by re-tagging and pushing to test/prod

podman tag myapp:1.0 quay.apps.uconn.edu/<org>/test:myapp_v1.0

podman push quay.apps.uconn.edu/<org>/test:myapp_v1.0

podman tag myapp:1.0 quay.apps.uconn.edu/<org>/prod:myapp_v1.0

podman push quay.apps.uconn.edu/<org>/prod:myapp_v1.0
```

Why this pattern: it keeps repository structure consistent (dev/test/prod) and makes promotion/auditing easier.

7) Pull an image (local)

```
podman pull quay.apps.uconn.edu/<org>/dev:myapp_v1.0
```

8) Using images in OpenShift

Deployments can reference Quay images directly, for example:

```
image: quay.apps.uconn.edu/<org>/dev:myapp_v1.0
```

Cluster pulls require an imagePullSecret attached to your ServiceAccount. Our onboarding automation typically creates and attaches this for the standard ServiceAccounts (default, builder, deployer) in your dev/test/prod namespaces.

9) How to find your imagePullSecret in OpenShift

In your namespace (example: <team>-dev), check the ServiceAccount:

```
oc -n <team>-dev get sa default -o
jsonpath='{.imagePullSecrets[*].name}{"\n"}'
```

```
oc -n <team>-dev describe sa default | egrep -i 'Image pull secrets|quay-pull'
```

Look for a secret named like: quay-pull-<team>. If it's missing, contact Platform Engineering.

10) Best practices

- Use versioned tags (e.g., 1.0.3) rather than only latest.
- Keep repositories private unless there is a documented need to make them public.
- Do not store large base OS images unless required for your app.
- Treat robot tokens like passwords; store them in approved secret stores (not in git).

Need help?

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